

Hyunsoo Kim

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EDUCATION

The University of Texas at Austin

Austin, TX

Ph.D. in Electrical and Computer Engineering

Sep 2026 – Present

- Co-advised by Prof. Guandao Yang and Prof. Diana Marculescu
- Research: Leveraging image and video generation models for visual understanding and perception

Korea University

Seoul, Republic of Korea

M.S. in Artificial Intelligence

Sep 2024 – Aug 2026

- Co-advised by Prof. Donghyun Kim and Prof. Suhyun Kim
- Research: Diffusion models, data-free synthesis, AI safety, and generative model alignment

Inha University

Incheon, Republic of Korea

B.S. in Statistics and Software Engineering

Mar 2018 – Aug 2024

- Two-year break for military service (Jul 2019 – Feb 2021)

RESEARCH INTERESTS

Video generation and world models (autoregressive video generation, video-based world models)

Generate to Understand (leveraging generative models for perception and visual understanding)

PUBLICATIONS

- [C5] **Consistency-Preserving Concept Erasure via Unsafe-Safe Pairing (PAIR)** NeurIPS 2026
- Yongwoo Kim*, Sungmin Cha*, **Hyunsoo Kim**, Jaewon Lee, Donghyun Kim (*: equal first authorship)
 - Proposed PAIR, a consistency-preserving concept erasure framework that reframes erasure as semantic realignment using unsafe-safe pairs, with Fisher-weighted DoRA initialization for selective suppression while maintaining generation quality
- [C4] **Correlation-Weighted Multi-Reward Optimization for Compositional Generation** ECCV 2026
- Jungmyung Wi, **Hyunsoo Kim**, Donghyun Kim
 - Proposed correlation-based reward reweighting for multi-concept text-to-image generation; dynamically estimates which concepts conflict across generated samples and assigns them stronger optimization signals via GRPO fine-tuning on SD3.5 and FLUX.1-dev
- [C3] **Self-Improving Diffusion Classifiers with Minority Preference Optimization** ECCV 2026
- **Hyunsoo Kim***, Jungmyung Wi*, Soobin Um, Donghyun Kim[†], Suhyun Kim[†] (*: equal contribution, †: equal advising)
 - Introduced MiPO, a self-improving framework that expands diffusion model coverage in low-density regions via GRPO with a novel DDIM-reconstruction-based minority preference reward
 - Improved zero-shot diffusion classification by up to 3.8% across multiple benchmarks without any additional real image data
- [C2] **Diffusion-assisted Data-free Image Synthesis with Domain-Class Alignment** ICML 2025
- Yujin Kim*, **Hyunsoo Kim***, Hyunwoo J. Kim, Suhyun Kim (*: equal contribution)
 - Proposed DDIS, a framework leveraging text-to-image diffusion models for data-free knowledge distillation with explicit alignment of domain and class distributions
- [C1] **Difference Inversion: Interpolate and Isolate the Difference with Token Consistency** CVPR 2025
- **Hyunsoo Kim**, Donghyun Kim[†], Suhyun Kim[†] (†: equal advising)
 - Proposed a token-consistency-based DDIM inversion method that interpolates and isolates visual differences between image pairs, enabling image analogy generation for style transfer, object manipulation, and scene editing

[P1] Custom Forcing: Training-Free Subject Customization for AR Video Generation

Preprint

- Yunseung Ok*, **Hyunsoo Kim***, Minseo Kim, Suhyun Kim[†] (*: equal contribution, †: corresponding author)
- Proposed a training-free method that customizes streaming autoregressive video models by writing reference images into the KV cache, with drift-adaptive value amplification and anchor contrast guidance to keep the subject identity over two-minute rollouts
- Kept DINO-I between 0.58 and 0.62 over two minutes while fixed anchors fell to 0.42, generating each frame 9.5–28.5× faster than causal image- and reference-to-video baselines

TEACHING

Teaching Assistant, ECE 381V: Advanced Topics in Computer Vision, The University of Texas at Austin Fall 2026

EXPERIENCE

Korea Institute of Science and Technology (KIST)

Seoul, Republic of Korea

*Graduate Researcher (Advisor: Suhyun Kim)**Sep 2024 – Aug 2026*

- Led research on image analogy generation, safe image translation, and self-improving diffusion classifiers
- Published 2 first-author papers at ECCV 2026 and CVPR 2025; 1 co-first-author paper at ICML 2025
- Developed training pipelines for diffusion model fine-tuning with RLHF/GRPO on multi-GPU clusters

Korea Institute of Science and Technology (KIST)

Seoul, Republic of Korea

*Undergraduate Research Intern (Advisor: Suhyun Kim)**Mar 2024 – Jun 2024*

- Researched data-free image synthesis by optimizing text embeddings on text-to-image diffusion models
- Contributed to DDIS framework that was later accepted at ICML 2025

Medical Image Innovation Lab, Seoul National University

Seoul, Republic of Korea

*Undergraduate Intern (Advisor: Wonjin Lee)**Dec 2023 – Feb 2024*

- Researched conditional generation and inpainting in medical imaging using diffusion models
- Implemented and evaluated diffusion-based inpainting pipelines for CT and MRI reconstruction tasks